Computations with simplicial tensor modules

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Abstract

We compute the homology of simplicial modules generated by hypermatrices over a field. Faces delete the same index along every axis, and degeneracies duplicate it. The Python package *SimplicialTensors* implements these operations. Projecting finitely many subtensors to normalized chains reduces homology to linear algebra. For fixed generator degree, every isomorphism of quotient complexes lifts to an automorphism of the normalized simplex complex. Ordinary cycle and boundary dimensions determine which kernels are realized in the tensor module. A 3×3 family exhibits a homology jump, and one-entry tensors realize every positive homology degree. Three-axis tensors realize the incidence complex of any finite graph. Over an infinite field, generic kernel families have rational image closures in products of Grassmannians. The full module associated to a shape is contractible. With k axes, its horn fillers are unique precisely above degree k. On rooted trees the omitted-face map splits onto its image, which can be a proper submodule of the boundary kernel. Every horn on a tree with at least $k+2$ vertices fills uniquely. For $k \geq 2$, a tree with $k+1$ vertices has an inner horn with $k!$ independent filling obstructions.

1 Introduction

Normalized chains of the simplicial module generated by a hypermatrix can be computed by finite linear algebra. For a matrix, the face d_i deletes row i and column i, and the degeneracy s_i duplicates both. For a hypermatrix with more axes, delete or duplicate index i along every axis. In degree n, these operations act on the initial block of indices $0, \ldots, n$ on every axis and carry along any indices after that block. The alternating sum of the faces gives the chain differential.

Starting with a hypermatrix in degree n over a field K, take the linear spans of its images under faces and degeneracies. To compute the normalized chains in degree p, retain each choice of $p+1$ indices from the initial block, using the same choice on every axis, and project the resulting hypermatrices to normalized chains. They span the degree-p chain group. There are no normalized chains above degree n, so the homology follows from finite linear algebra (Section 6).

For a fixed generator degree, the normalized kernels classify modules with a chosen generator. Forgetting that choice amounts to taking automorphism orbits of the normalized simplex complex: every isomorphism between its quotients lifts to an automorphism (Proposition 6.1). A kernel is realized in the tensor module precisely when its quotient's ordinary cycle and boundary dimensions fit those of the ambient normalized complex (Proposition 6.2).

The *SimplicialTensors* repository supplies the simplicial operations and examples. The module generated by the 2×2 matrix with a single 1 in position $(0, 1)$ has one-dimensional first homology (Example 2.1). A one-parameter 3×3 family has one-dimensional first homology except at a single acyclic member (Example 6.3). More generally, for each $q \geq 1$, a coordinate tensor with $q+1$ axes generates the normalized complex $K[q]$ (Example 6.4). Encoding the vertices and edges of a graph in matrix slices of hypermatrices with three axes gives the graph's incidence complex (Proposition 6.5). For the Petersen graph the computation gives first Betti number six.

A coordinate basis gives the full module's horn kernels and disk decomposition (Theorem 3.2 and Proposition 4.1). Section 7 computes omitted-face lifts and a sharp horn-filling bound on rooted trees.

Appendix A parameterizes generic normalized kernels in Grassmannians and proves rationality of their image closure.

2 The construction

Throughout, A is a nonzero commutative ring with identity, $k \geq 1$, and $[p] = \{0, \ldots, p\}$. Let Δ denote the category of finite nonempty ordinals and nondecreasing morphisms.

Fix a shape $\vec{s} = (n_1, \ldots, n_k) \in \mathbb{Z}^k_{>0}$ and put

$$n = \min_a n_a - 1, \qquad c_a = n_a - 1 - n, \qquad I_p = \prod_{a=1}^{k} [p + c_a].$$

The integers $c_a = n_a - \min_b n_b$ are the *shape offsets*. The offsets are nonnegative and at least one is zero. They determine the simplicial module, and n is the degree in which the original shape occurs. We allow arbitrary nonnegative tail lengths $\mathbf{c} = (c_1, \ldots, c_k)$ and use $X(\mathbf{c}; A)$ for the resulting module. A common rank shift $r \geq 0$ replaces the offsets of a shape by $(c_a + r)_a$.

For $c \geq 0$, put $J_c([p]) = [p + c]$. For $\alpha : [p] \to [q]$, define

$$J_c(\alpha)(r) = \begin{cases} \alpha(r), & 0 \leq r \leq p, \\ q + t, & r = p + t, \quad 1 \leq t \leq c. \end{cases}$$

This defines a covariant functor $J_c : \Delta \to \mathbf{Set}$, since composition acts separately on the initial ordinal and the final c elements. Consequently $I = \prod_a J_{c_a}$ is cosimplicial and

$$X_p = X_p(\mathbf{c}; A) = A^{I_p}, \qquad X(\alpha)(T) = T \circ I(\alpha)$$

defines a functor $X : \Delta^{\mathrm{op}} \to \mathbf{Mod}_A$. This is the *diagonal simplicial tensor module* with tail lengths $\mathbf{c}$. For the offsets of a shape $\vec{s}$, write $X(\vec{s}; A)$ for this same module. Write E_m for the indicator function of $m \in I_p$.

For $p \geq 1$ and $0 \leq i \leq p$, the index coface $\Delta_i : I_{p-1} \to I_p$ inserts a gap at i in every coordinate; its image consists of the tuples avoiding i. For $p \geq 0$ and $0 \leq i \leq p$, the codegeneracy $\Sigma_i : I_{p+1} \to I_p$ identifies i and $i + 1$ in every coordinate. Thus $d_i T = T \circ \Delta_i$ and $s_i T = T \circ \Sigma_i$. In particular,

$$d_i E_m = \begin{cases} 0, & i \in \mathrm{im}(m), \\ E_{\Delta_i^{-1} m}, & i \notin \mathrm{im}(m), \end{cases} \qquad \mathrm{im}(m) = \{m_1, \ldots, m_k\}. \tag{1}$$

The nonzero images in this formula are distinct basis elements. There are no outgoing face maps in degree zero. The associated chain complex has $X_{-1} = 0$ and differential

$$\partial_p = \sum_{i=0}^{p} (-1)^i d_i \quad (p \geq 1), \qquad \partial_0 = 0.$$

For a simplicial module Y, we write $H_p(Y)$ for the homology of this unaugmented complex. For instance, the faces of the vector $(3, 5)$ are (5) and (3), so its boundary is (2).

The module X is the diagonal of the k-fold simplicial module with term $A^{\prod_a [p_a + c_a]}$ in multidegree $(p_1, \ldots, p_k)$. Compare Quillen's diagonal of a double semisimplicial group [8]. The finite coordinate bases also identify $X(\mathbf{c}; A) \otimes_A B$ with $X(\mathbf{c}; B)$ for any ring homomorphism $A \to B$.

For $T \in X_n$, its generated simplicial submodule has degree-p part

$$\langle T \rangle_p = \mathrm{span}_A \{X(\alpha)(T) : \alpha : [p] \to [n]\}.$$

This is closed under faces and degeneracies by functoriality.

Example 2.1. Over a field K, take shape $(2,2)$, write $E_{ij} = E_{(i,j)}$ and let

$$T = E_{01} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad Y = \langle T \rangle \subseteq X.$$

Both faces vanish, so $Y_0 = 0$, $Y_1 = KT$, and

$$Y_2 = \operatorname{span}_K\{s_0T, s_1T\}, \qquad s_0T = E_{02} + E_{12}, \quad s_1T = E_{01} + E_{02}.$$

Both degeneracies have zero boundary by the simplicial identities. Thus $H_1(Y) \cong K$. In the ambient complex, $\partial_2 E_{12} = T$. The tensor E_{12} lies outside Y_2 because the two displayed generators have coordinates satisfying $x_{02} = x_{12} + x_{01}$.

The accompanying script `dstm_examples.py` reproduces Examples 2.1 and 3.3 using *Simplicial-Tensors* [5]. It computes the first homology dimension over $\mathbb{Q}$ and the two free horn coordinates.

3 Horn kernels and normalization

For any simplicial A-module Y, put

$$N_p(Y) = \bigcap_{i=1}^{p} \ker d_i, \qquad D_p(Y) = \sum_{i=0}^{p-1} \operatorname{im} s_i.$$

Here $N_0(Y) = Y_0$, $D_0(Y) = 0$, and the normalized differential is d_0 in positive degrees and zero in degree zero; this is the convention in [13, Exercise 8.3.4]. The normalization theorem gives $Y_p = N_p(Y) \oplus D_p(Y)$ [2, Corollary 1.12]. The Dold–Kan theorem gives an equivalence with non-negative chain complexes [13, Theorem 8.4.1].

For $p \geq 1$ and $0 \leq j \leq p$, let $R_{p,j}(Y) = \bigcap_{0 \leq i \leq p,\ i \neq j} \ker d_i$. The module $\Lambda_{p,j}(Y)$ of compatible horns consists of tuples $(y_i)_{i \neq j}$ with $y_i \in Y_{p-1}$ satisfying $d_i y_\ell = d_{\ell-1} y_i$ for $i < \ell$, $i, \ell \neq j$; for $p = 1$ there are no compatibility conditions. The horn map is $\Phi_j : Y_p \to \Lambda_{p,j}(Y)$, $y \mapsto (d_i y)_{i \neq j}$. Every simplicial module is Kan, so Φ_j is surjective [13, Lemma 8.2.8]. Its fibers are torsors for $R_{p,j}(Y)$.

Lemma 3.1. *For every simplicial A-module Y, $p \geq 1$ and $0 \leq j \leq p$,*

$$Y_p = R_{p,j}(Y) \oplus D_p(Y).$$

The restriction $\Phi_j|_{D_p(Y)}$ is an isomorphism. Thus every horn has a unique filler in the submodule $D_p(Y)$.

Proof. For $0 \leq j < p$, the simplicial identities give inverse maps

$$F_j = 1 - s_j d_j : R_{p,j}(Y) \longrightarrow R_{p,j+1}(Y), \qquad G_j = 1 - s_j d_{j+1} : R_{p,j+1}(Y) \longrightarrow R_{p,j}(Y).$$

The faces of $s_j d_j x$ for $x \in R_{p,j}(Y)$ vanish away from $j, j+1$. For $i < j$ use $d_i d_j = d_{j-1} d_i$, and for $i > j+1$ use $d_{i-1} d_j = d_j d_i$ after moving d_i past s_j. The identities $d_j s_j = d_{j+1} s_j = 1$ give $d_j F_j x = 0$ and $d_{j+1} F_j x = -d_j x$, so $G_j F_j x = x$. For $y \in R_{p,j+1}(Y)$ the same face identities applied to $s_j d_{j+1} y$ give $d_{j+1} G_j y = 0$ and $d_j G_j y = -d_{j+1} y$, so $F_j G_j y = y$. Both maps induce the identity modulo $D_p(Y)$. Since $R_{p,0}(Y) = N_p(Y)$ maps isomorphically onto $Y_p / D_p(Y)$, the same holds for every $R_{p,j}(Y)$. This proves the direct sum. The final assertion follows from surjectivity of Φ_j and $\ker \Phi_j = R_{p,j}(Y)$. □

For simplicial objects in finite-dimensional real vector spaces the horn-space isomorphism is also proved in [9, Lemma 2.10]; Lemma 3.1 gives an argument over arbitrary A.

Theorem 3.2. *Let $X = X(\mathbf{c}; A)$, $p \geq 1$, and $0 \leq j \leq p$. Put $F_j = [p] \setminus \{j\}$. Then*

$$R_{p,j}(X) = \bigoplus_{\substack{m \in I_p \\ F_j \subseteq \mathrm{im}(m)}} AE_m, \qquad \operatorname{rank} R_{p,j}(X) = \sum_{t=0}^{p} (-1)^t \binom{p}{t} \prod_{a=1}^{k} (p + c_a + 1 - t). \tag{2}$$

In particular, $R_{p,j}(X) \neq 0$ if and only if $p \leq k$. Thus horn fillers are nonunique for $1 \leq p \leq k$ and unique for $p > k$, independently of j and of the offsets.

Proof. By (1), $d_i T = 0$ exactly when T is supported on tuples containing i. Intersecting these support conditions gives (2). Inclusion–exclusion counts tuples containing all p elements of F_j: forbidding a specified t of these elements leaves $p + c_a + 1 - t$ choices in coordinate a. A tuple of length k cannot contain $p > k$ distinct values. If $p \leq k$, assign the p required values to distinct coordinates and fill the remaining coordinates arbitrarily in $[p]$. This gives a basis element, nonzero because $A \neq 0$. □

Example 3.3. For shape $(3, 3)$, consider the 2-horn with missing face 0 and prescribed faces

$$d_1 T = \begin{pmatrix} 1 & 3 \\ 7 & 9 \end{pmatrix}, \qquad d_2 T = \begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix}.$$

Their top left entries agree, so the faces are compatible. Every filler is of the form

$$T = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & u \\ 7 & v & 9 \end{pmatrix}, \qquad u, v \in A.$$

Thus the difference of two fillers lies in $R_{2,0}(X) = AE_{(1,2)} \oplus AE_{(2,1)}$, whose basis consists of the two index pairs containing both 1 and 2.

An algebraic r-hypergroupoid is a simplicial module with unique horn fillers in dimensions greater than r [3, 4]. By Theorem 3.2, X is such an object exactly when $r \geq k$. For the original degree $n = \min_a n_a - 1$, the n-hypergroupoid condition is $k \leq n$, with nonunique n-horn fillers when $k = n \geq 1$.

4 The normalized complex

Write $B = A^{\prod_a C_a}$, where C_a is the set of c_a tail labels on axis a. At degree p the tail labels are $p + 1, \ldots, p + c_a$. Projection onto tuples of tail labels defines a simplicial morphism $X \to \underline{B}$, with section given by extension by zero. Write $Z_0 = \ker(X_0 \to B)$ and $Z_p = \ker(d_0 : N_p(X) \to N_{p-1}(X))$ for $p \geq 1$. Then

$$Z_p = \bigoplus_{\substack{m \in I_p \\ [p] \subseteq \mathrm{im}(m)}} AE_m \qquad (p \geq 0). \tag{3}$$

Proposition 4.1. *Let $X = X(\mathbf{c}; A)$. For $p \geq 1$ and $0 \leq j \leq p$, the extension by zero $\iota_j : Z_{p-1} \to R_{p,j}(X)$ along Δ_j gives $R_{p,j}(X) = Z_p \oplus \iota_j Z_{p-1}$ and $d_j \iota_j = 1$. If $D^r(M)$ denotes the identity complex $M \xrightarrow{1} M$ in degrees $r, r-1$, then*

$$N_\bullet(X) \cong B[0] \oplus \bigoplus_{p=0}^{k-1} D^{p+1}(Z_p). \tag{4}$$

In particular $H_0(X) \cong A^{\prod_a c_a}$ and $H_p(X) = 0$ for $p > 0$.

Proof. Partition the basis of $R_{p,j}(X)$ according as the tuple contains j or avoids it. The tuples containing j give the basis of Z_p. Deleting the gap at j sends the other tuples bijectively to the basis of Z_{p-1}, and d_j induces this bijection by (1). For $j = 0$ this is the normalized differential. In degree zero $X_0 = B \oplus Z_0$. Finally $Z_p = 0$ for $p \geq k$, since a tuple of length k cannot contain $p+1$ distinct indices. □

The contraction on the disk summands sends $E_m \in Z_p$ to $E_{m+\mathbf{1}} \in N_{p+1}(X)$ and vanishes on the other summands. The coordinate bases show that normalized chains, cycles and horn kernels commute with arbitrary base change. In degree zero the cycles are $B \oplus Z_0$.

Corollary 4.2. *Let $b_p = \operatorname{rank} Z_p$ and $b_{-1} = 0$. Then*

$$b_p = \sum_{t=0}^{p+1} (-1)^t \binom{p+1}{t} \prod_{a=1}^{k} (p + c_a + 1 - t),$$
$$\operatorname{rank} N_p(X) = b_p + b_{p-1} + \mathbf{1}_{p=0} \prod_a c_a.$$

For $c_a = 0$ on all axes, $b_p = (p+1)!\left\{ {k \atop p+1} \right\}$, where $\left\{ {k \atop r} \right\}$ is a Stirling number of the second kind.

Proof. Apply inclusion–exclusion to (3) and use (4). With all tails empty, tuples covering $[p]$ are surjections from a k-element set onto $[p]$ [10, §1.4]. □

Example 4.3. For square matrices with empty tails, $N_\bullet(X) \cong D^1(A) \oplus D^2(A^2)$. The normalized ranks are $1, 3, 2$ in degrees $0, 1, 2$. The basis elements $E_{(1,2)}$ and $E_{(2,1)}$ in degree 2 map to $E_{(0,1)}$ and $E_{(1,0)}$ under d_0.

Corollary 4.4. *Over a fixed nonzero commutative ring, two modules $X(\mathbf{c}; A)$ are isomorphic precisely when their multisets of tail lengths agree. For modules specified by shapes, this means permuting the entries and adding the same integer to each entry.*

Proof. An isomorphism equates the ranks in every degree. Hence the polynomials $\prod_a (x + c_a + 1)$ agree on all nonnegative integers, so their roots agree with multiplicities. A permutation matching the tails induces a simplicial isomorphism on coordinates. □

5 Split contractions and rank shifts

For A-modules E, W, put $V_A(E, W)_p = E^{p+1} \oplus W$. Faces delete and degeneracies repeat E-coordinates, fixing W. This is the Čech nerve of $E \oplus W \to W$ [11, Tag 019P]. On $V = \bigotimes_a V_A(E_a, W_a)$ the split projections give simplicial morphisms $\epsilon : V \to \underline{B}$ and $j : \underline{B} \to V$, where $B = \bigotimes_a W_a$. Taking $E_a = A$ and $W_a = A^{c_a}$ recovers $X(\mathbf{c}; A)$.

Let $H_p : V_p \to V_{p+1}$ be the tensor product of the maps $(e_0, \ldots, e_p, w) \mapsto (0, e_0, \ldots, e_p, w)$. In degree -1 take $H_{-1} = j_0 : B \to V_0$. Then $d_0 H_p = 1$ and $d_i H_p = H_{p-1} d_{i-1}$ for $1 \leq i \leq p+1$, using $d_0 = \epsilon_0$ on V_0 in the latter identity. Summing with signs gives $\partial_{p+1} H_p + H_{p-1} \partial_p = 1$ for $p \geq 1$ and $\partial_1 H_0 = 1 - j_0 \epsilon_0$. Hence the unaugmented complex deformation retracts onto $B[0]$. Projecting H to the normalized summand gives the same retraction on $N(V)$. For the offsets of a shape, some $c_a = 0$, so $B = 0$ and the full complex is contractible.

For coordinate tensors, $H_p(E_m) = E_{m+\mathbf{1}}$. Both this contraction and the normalized contraction commute with axis permutations preserving the tails, so they restrict to the symmetric and sign eigenspaces. For square complex matrices they commute with conjugate transpose on the underlying real complexes. If some tail is empty, the filtration spanned by E_m with $\min_a m_a \geq p + t$, indexed

by $t \in \mathbb{Z}$, is decreasing and finite in each degree. A surviving face lowers the minimum by at most one, and H_p raises it by one. The induced contraction on the associated graded complex gives zero homology.

For the two-variable form of the rank shift, let Δ_+ be Δ with the empty ordinal $[-1]$ adjoined. For $p, q \geq -1$ use the index set

$$I_{p,q} = \prod_{a=1}^{k} \big(([p] \oplus [c_a - 1]) \oplus [q]\big),$$

where $\oplus$ denotes ordinal sum. On a pair of morphisms $(\alpha, \beta) : ([p],[q]) \to ([p'],[q'])$, each factor uses $\alpha \oplus 1_{[c_a-1]} \oplus \beta$. Pullback gives a functor $(\Delta_+ \times \Delta_+)^{\mathrm{op}} \to \mathbf{Mod}_A$. Its slice at q is the horizontally augmented module with tail lengths $c_a + q + 1$. Thus $r = q + 1$, and $q = -1$ gives $r = 0$. Moving the fixed block after $[q]$ identifies this bifunctor with the pullback of the augmented X along ordinal sum. The degree-zero faces into degree -1 restrict to the surviving blocks. Horizontal and vertical maps commute because they act on separate blocks. For the alternating face boundaries, the total differential on bidegree (p, q) is $\partial_h + (-1)^p \partial_v$. The horizontal contraction prepends zero on the first block and commutes with all vertical maps. It contracts the total complex when the horizontal augmentation is included.

6 Generated submodules

Let K be a field, fix $n \geq 0$ and nonnegative tail lengths $\mathbf{c} = (c_1, \ldots, c_k)$, and put $X = X(\mathbf{c}; K)$. Let $P = K[\Delta^n]$ be the free simplicial K-module on the standard n-simplex, so P_p has basis the nondecreasing maps $[p] \to [n]$. For $T \in X_n$, its representing property gives

$$f_T : P_\bullet \longrightarrow X_\bullet, \qquad f_T(\alpha) = X(\alpha)(T) \quad \big(\alpha : [p] \to [n]\big).$$

Its image is the simplicial submodule $\langle T \rangle$ generated by T, and $P/\ker f_T \cong \langle T \rangle$. Put

$$C = N(P), \qquad g_T = N(f_T), \qquad L(T) = \ker g_T.$$

Normalization is exact, so

$$N(\langle T \rangle) \cong C/L(T). \tag{5}$$

Here C_p has dimension $\binom{n+1}{p+1}$ for $0 \leq p \leq n$ and is zero otherwise. Hence the finitely many spaces $L_p(T)$ determine the entire simplicial kernel by Dold–Kan. Equality of these kernels is equivalent to an isomorphism of the generated modules sending one distinguished simplex to the other. For instance, E_{01} and E_{10} have the same kernel in shape $(2, 2)$ and generate distinct embedded submodules.

Homology is computed from normalized complexes. Since $H_0(C) = K$ and $H_j(C) = 0$ for $j > 0$, the short exact sequence $0 \to L(T) \to C \to C/L(T) \to 0$ gives

$$\begin{aligned} H_j(\langle T \rangle) &\cong H_{j-1}(L(T)) \qquad (j \geq 2), \\ 0 \longrightarrow H_1(\langle T \rangle) &\longrightarrow H_0(L(T)) \xrightarrow{\epsilon} K \longrightarrow H_0(\langle T \rangle) \longrightarrow 0, \end{aligned} \tag{6}$$

where ϵ is induced by the augmentation sending every vertex of Δ^n to 1.

6.1 Quotients and automorphisms

We work in the category $\mathrm{Ch}_{\geq 0}(K)$ of nonnegative chain complexes of K-vector spaces and chain maps. A complex is finite if the direct sum of its chain groups is finite-dimensional. For a complex E, write

$$Z_p^{\mathrm{ord}}(E) = \ker(\partial_p : E_p \to E_{p-1}), \qquad B_p(E) = \mathrm{im}(\partial_{p+1} : E_{p+1} \to E_p), \qquad E_{-1} = 0.$$

Thus $Z_0^{\mathrm{ord}}(E) = E_0$. For a K-vector space V and $p \geq 0$, let $S^p(V)$ be the complex with V in degree p and zero elsewhere. Choosing complements of boundaries inside cycles, and of cycles inside each chain group, gives

$$E \cong \bigoplus_{p\geq 0} S^p(H_p(E)) \oplus \bigoplus_{p\geq 1} D^p(B_{p-1}(E)) \tag{7}$$

for every finite complex E. Indeed, the differential maps the chosen complement of $Z_p^{\mathrm{ord}}(E)$ isomorphically onto $B_{p-1}(E)$. These choices also show that homology dimensions and differential ranks determine the chain isomorphism class and admit cancellation under direct sums.

For the normalized simplex complex, the dimensions of its chain groups and its homology give

$$C \cong S^0(K) \oplus \bigoplus_{p=1}^{n} D^p\big(K^{\binom{n}{p}}\big). \tag{8}$$

Both $S^0(V)$ and $D^p(V)$, for $p \geq 1$, are projective objects: a chain map from either is determined by an arbitrary linear map from V into the target's degree-zero or degree-p group, respectively. Evaluation in either degree and $\mathrm{Hom}_K(V, -)$ are exact. Consequently C is projective.

Proposition 6.1. *Fix a field K and $n \geq 0$, and let $C = N(K[\Delta^n])$. For subcomplexes $L, M \subseteq C$, every specified chain isomorphism $\varphi : C/L \to C/M$ lifts to a chain automorphism $u : C \to C$ making the following square commute:*

$$\begin{array}{ccc} C & \xrightarrow{\ u\ } & C \\ {\scriptstyle \pi_L}\downarrow & & \downarrow{\scriptstyle \pi_M} \\ C/L & \xrightarrow{\ \varphi\ } & C/M. \end{array}$$

Here π_L and π_M are the quotient maps. In particular, $C/L \cong C/M$ if and only if $u(L) = M$ for some $u \in \mathrm{Aut}_{\mathrm{Ch}_{\geq 0}(K)}(C)$. The orbits of this automorphism group on subcomplexes of C classify the chain isomorphism classes of quotients of C.

Proof. Projectivity supplies chain maps $f, g : C \to C$ with $\pi_M f = \varphi \pi_L$ and $\pi_L g = \varphi^{-1} \pi_M$. Put $h = gf$, so $\pi_L h = \pi_L$. Choose m large enough that the images and kernels of h^m stabilize in every degree. The Fitting decomposition is a direct sum of subcomplexes

$$C = U \oplus V, \qquad U = \operatorname{im} h^m, \quad V = \ker h^m,$$

with $h_U = h|_U$ invertible. Since $\pi_L h^m = \pi_L$, we have $V \subseteq L$. Let $e : C \to U$ be the projection and define $r = h_U^{-1} e g : C \to U$. Then $rf|_U = 1_U$, so

$$C = f(U) \oplus V', \qquad V' = \ker r, \qquad f|_U : U \xrightarrow{\sim} f(U).$$

Write $\lambda = \pi_L|_U : U \to C/L$. Since $V \subseteq L$ and $\pi_L h = \pi_L$, we have $\lambda e = \pi_L$ and $\lambda h_U^{-1} = \lambda$. Hence $\lambda r = \pi_L g$ and $\pi_M = \varphi \pi_L g = \varphi \lambda r$, so $V' \subseteq M$.

Cancellation in (7) gives a chain isomorphism $a : V \to V'$. Define u by $u|_U = f|_U$ and $u|_V = a$. The two direct sums show that u is invertible. Its quotient map agrees with $\varphi \pi_L$ on U and vanishes on V, proving the specified lifting identity. Taking kernels gives $u(L) = M$. Conversely, such an automorphism induces an isomorphism of quotients. □

For $a \in \mathrm{Aut}_{\mathrm{Ch}_{\geq 0}(K)}(C)$, let $\widetilde{a} : P \to P$ be its simplicial automorphism under Dold–Kan. The tensor

$$T' = f_{T,n}\big(\widetilde{a}_n(\mathrm{id}_{[n]})\big) \in X_n$$

satisfies $f_{T'} = f_T\widetilde{a}$ by the representing property of P. Hence $L(T') = a^{-1}L(T)$ and $\langle T'\rangle = \langle T\rangle$. Thus the set $\{L(T) : T \in X_n\}$ is invariant under the automorphism group. Equation (5), Dold–Kan and Proposition 6.1 identify its orbits with the abstract simplicial isomorphism classes of the generated modules. There are only finitely many such orbits: the chain dimensions and differential ranks of C/L are bounded integers.

6.2 Realizable kernels

Put $\mathcal{D} = N(X(\mathbf{c}; K))$. A subcomplex $L \subseteq C$ is *realizable in* X if $L = L(T)$ for some $T \in X_n$.

Proposition 6.2. *Let K be a field, $n \geq 0$, $C = N(K[\Delta^n])$, and $X = X(\mathbf{c}; K)$ for arbitrary nonnegative tail lengths $\mathbf{c}$. For a subcomplex $L \subseteq C$ and $Q = C/L$, the following are equivalent:*

1. *L is realizable in X.*
2. *There is a chain monomorphism $Q \hookrightarrow \mathcal{D} = N(X)$.*
3. *For every $p \geq 0$,*

$$\dim_K Z_p^{\mathrm{ord}}(Q) \leq \dim_K Z_p^{\mathrm{ord}}(\mathcal{D}), \qquad \dim_K B_p(Q) \leq \dim_K B_p(\mathcal{D}). \tag{9}$$

Proof. A realizing map $g_T : C \to \mathcal{D}$ induces the required monomorphism $C/L(T) \to \mathcal{D}$. Any chain monomorphism restricts to injections on ordinary cycles and on boundaries, proving the inequalities.

Conversely, assume (9). Basis extension gives, in each degree, an injection

$$j_p : Z_p^{\mathrm{ord}}(Q) \longrightarrow Z_p^{\mathrm{ord}}(\mathcal{D}), \qquad j_p(B_p(Q)) \subseteq B_p(\mathcal{D}).$$

Choose $Q_p = Z_p^{\mathrm{ord}}(Q) \oplus W_p$. For $p > 0$, $\partial_Q : W_p \to B_{p-1}(Q)$ is an isomorphism. Lift $j_{p-1}\partial_Q|_{W_p}$ through $\partial_{\mathcal{D}} : \mathcal{D}_p \to B_{p-1}(\mathcal{D})$ to a linear map $\ell_p : W_p \to \mathcal{D}_p$. Such a lift exists because its image lies in the indicated boundary space. Set $i_p(z+w) = j_p(z) + \ell_p(w)$; in degree zero use $i_0 = j_0$, since $W_0 = 0$. These maps form a chain map. For $p > 0$, if $i_p(z+w) = 0$, applying the differential gives $j_{p-1}\partial_Q w = 0$, hence $w = 0$ and then $z = 0$. The degree-zero map is injective by construction. Thus $i : Q \to \mathcal{D}$ is a monomorphism.

The composite $C \to Q \xrightarrow{i} \mathcal{D}$ corresponds under Dold–Kan to a simplicial map $P = K[\Delta^n] \to X$. The representing property of P identifies it with f_T for the image T of the identity n-simplex. Its normalized kernel is L, so $L = L(T)$. □

Proposition 4.1 identifies $B_p(\mathcal{D})$ with the coordinate module Z_p of Section 4 for every $p \geq 0$. For $p > 0$, $Z_p^{\mathrm{ord}}(\mathcal{D}) = Z_p$, whereas

$$Z_0^{\mathrm{ord}}(\mathcal{D}) = B \oplus Z_0, \qquad \dim_K Z_0^{\mathrm{ord}}(\mathcal{D}) = \prod_a (c_a + 1), \qquad \dim_K B_0(\mathcal{D}) = \prod_a (c_a + 1) - \prod_a c_a.$$

In particular, positive tails contribute ordinary degree-zero cycles through the summand B. The remaining dimensions in (9) are given by Corollary 4.2.

For example, when $\mathbf{c} = (1, 1)$, the ordinary cycle dimensions of $\mathcal{D}$ are $(4, 2, 0)$ and its boundary dimensions are $(3, 2, 0)$. The complex $Q = D^1(K^4)$ is a quotient of $N(K[\Delta^4])$ by (8). It satisfies the cycle inequalities, but $\dim B_0(Q) = 4 > 3 = \dim B_0(\mathcal{D})$, so it does not embed.

6.3 Matrix and sparse coordinate examples

Example 6.3. For $t \in K$, take empty tails and

$$T(t) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & t \end{pmatrix} \in X_2, \qquad Y_t = \langle T(t) \rangle.$$

In $N_1(X)$, use coordinates (E_{01}, E_{10}, E_{11}). The normalized restrictions to vertex pairs $01, 02, 12$ are

$$v_{01} = (1, 0, 1), \qquad v_{02} = (0, 1, t), \qquad v_{12} = (0, 0, t-1).$$

Their determinant is $t - 1$. The normalized image of the top simplex is $E_{12} - E_{21}$, whose differential is $E_{01} - E_{10}$. The differential out of $N_1(X)$ reads the E_{11} coordinate. Thus the differential ranks of the generated complex are $(0, 1, 1)$, indexed by source degree, and

	$(\dim N_0(Y_t), \dim N_1(Y_t), \dim N_2(Y_t))$	$(\dim H_0(Y_t), \dim H_1(Y_t), \dim H_2(Y_t))$
$t \neq 1$	$(1, 3, 1)$	$(0, 1, 0)$
$t = 1$	$(1, 2, 1)$	$(0, 0, 0)$.

For $t \neq 1$, the cycle $E_{01} = v_{01} - (t-1)^{-1} v_{12}$ represents the nonzero homology class. At $t = 1$, the cycle space in degree one is the line spanned by $v_{01} - v_{02} = E_{01} - E_{10}$, which is already a boundary.

In the vertex basis of $C_0 = K^3$, the degree-zero kernel is

$$L_0(T(t)) = \ker(0, 1, t) = \operatorname{span}_K\{(1, 0, 0), (0, -t, 1)\}.$$

These planes vary with t. For $t \neq 1$, the remaining normalized kernels are $L_1 = L_2 = 0$, and all the quotient complexes are isomorphic. Proposition 6.1 therefore places these distinct kernels in one automorphism orbit. At $t = 1$, L_1 is spanned by the normalized edge $[12]$ and $L_2 = 0$. For example, $T(4)$ has first Betti number one over $\mathbb{Q}$ and zero over $\mathbb{F}_3$.

Example 6.4. Fix $q \geq 1$, use $k = q + 1$ axes and any nonnegative tail lengths, and take $T_q = E_{(0,1,\ldots,q)} \in X_q(\mathbf{c}; K)$. Every face of T_q is zero, since its index tuple contains every active label. Hence T_q is normalized. All lower-degree generators of $\langle T_q \rangle$ vanish. In higher degrees, every nondecreasing map to $[q]$ is noninjective, so its image is a degeneracy of a lower-degree generator. Consequently

$$N(\langle T_q \rangle) = S^q(K), \qquad H_q(\langle T_q \rangle) \cong K, \qquad H_p(\langle T_q \rangle) = 0 \quad (p \neq q).$$

For $q = 2$ and empty tails this uses one nonzero entry in a $3 \times 3 \times 3$ tensor. For $q = 3$ it uses one entry in shape $(4, 4, 4, 4)$ and gives one-dimensional third homology. In the full module,

$$\partial_{q+1} E_{(1,2,\ldots,q+1)} = T_q.$$

Thus the same cycle is an ambient boundary. The number of axes is minimal: (3) gives $Z_q(X) = 0$ for $k \leq q$, so no simplicial submodule of such an X has nonzero qth homology.

The repository script `experiments/kernel_sequences.py` computes the kernel, quotient and homology data in these examples.

6.4 Graph chains in shifted matrix slices

Let a finite graph have v vertices and e oriented edges, each numbered separately starting at 1. Write $u_j \to w_j$ for edge j. For $r \geq \max(v, e)$ use $X((r, r, r); A)$. Write a tensor as a family of matrices, with its third coordinate specifying the slice. The following generators are supported in fixed tail slices:

$$a_t = E_{(0,t,t)} \in X_0, \qquad T_j = E_{(0,u_j+1,u_j+1)} + E_{(1,w_j+1,w_j+1)} + E_{(0,1,j+1)} \in X_1.$$

Here $1 \leq t \leq v$ and $1 \leq j \leq e$. Deleting index 1 gives $d_1 T_j = a_{u_j}$, and deleting index 0 gives $d_0 T_j = a_{w_j}$.

Proposition 6.5. *The simplicial submodule Y_G generated by the a_t and T_j has normalized complex $A^e \to A^v$ in degrees $1, 0$, with differential sending edge j to its target minus its source.*

Proof. The vertex tensors are distinct coordinate units. The normalized edges $b_j = T_j - s_0 a_{u_j}$ are linearly independent because the coordinate $(0, 1, j+1)$ occurs with coefficient 1 in b_j and vanishes on the other normalized edges. Also $d_0 b_j = a_{w_j} - a_{u_j}$. The other degree-one generators are degeneracies of vertices, and every generator in higher degrees is a degeneracy. The normalization projection kills these terms. □

The boundaries of the edges of a spanning forest form a basis of the kernel of the vertex-sum morphism on each component. Subtracting forest paths from the remaining edges gives a basis of cycles. Thus $H_0(Y_G) \cong A^b$ and $H_1(Y_G) \cong A^{e-v+b}$, where b is the number of components. The Petersen graph has ten vertices, fifteen edges, and one component, so its first homology is A^6. The script `rank_shifted_homology.py` computes its incidence rank from the normalized tensor faces over $\mathbb{Q}$ [6].

The two-axis cycle module in degree one has basis $E_{(0,1)}$ and $E_{(1,0)}$ by (3). Over a field, this bounds first homology of every two-axis simplicial submodule by dimension two.

7 Horns on rooted trees

Let Ω be the dendroidal category. Its objects are rooted trees and its morphisms are morphisms of the free symmetric coloured operads on those trees [7, §3]. For a tree T with edge set $E(T)$, put $I(T) = \prod_{a=1}^{k} (E(T) \sqcup C_a)$, where each C_a is a fixed finite set. Pullback along colour maps, extended by the identity on the C_a, defines $D : \Omega^{\mathrm{op}} \to \mathbf{Mod}_A$, with $D(T) = A^{I(T)}$. On linear trees the colour maps are the ordinal maps, so this restricts to $X(\mathbf{c}; A)$ for $c_a = |C_a|$.

An elementary inner face contracts an inner edge. For a tree with at least two vertices, an elementary outer face prunes an endpoint of the tree of vertices, retaining its one incident inner edge. The faces of a one-vertex tree are its individual edges. For a face $F : S \to T$, the horn omitting F is the union of the representables of the other elementary faces of T. Its data are tensors on those faces agreeing on common subfaces. This description follows by factoring dendroidal morphisms into degeneracies, isomorphisms and faces [1, Lemma 2.3]. Write $K_F(T)$ for the tensors that vanish on the included faces and $K(T)$ for those that vanish on all elementary faces. Identify face index sets with their images in $I(T)$.

Proposition 7.1. *Let $F : S \to T$ be an elementary face. The pullback $F^* : K_F(T) \to K(S)$ has kernel $K(T)$ and image*

$$A^{\,I(S) \setminus \bigcup_{G \neq F} I(S_G)} \subseteq K(S), \qquad G : S_G \to T.$$

The inclusion is by extension by zero in $I(S)$. The map onto this image splits by extension by zero in $I(T)$.

Proof. Each elementary face of S, composed with F, factors through another elementary face of T by the codimension-two face identity [1, Lemma 2.1]. Thus the restriction of $K_F(T)$ lies in $K(S)$. Membership in $K_F(T)$ means vanishing on the union of the included index sets. Restriction to $I(S)$ and extension by zero give the stated image and section. The kernel vanishes on F as well as on the included faces. □

Take the two-vertex tree with vertices $u : a \to e$ and $v : (e, b) \to r$, and omit the contraction of e. Its two included faces have colour sets $\{a, e\}$ and $\{e, b, r\}$, meeting in the single-edge tree on e. Every such inner horn fills by gluing the two arrays on their common index set. With two axes and empty tails, S is the corolla $(a, b) \to r$ and $K(S)$ is spanned by its six off-diagonal units. The omitted-face

map reaches the four ordered pairs joining a to b or r. The units at (b, r) and (r, b) are visible on the face with colours $\{e, b, r\}$ and have no lifts in $K_F(T)$.

Theorem 7.2. *For k axes and arbitrary fixed tails, all horns on trees with at least $k + 2$ vertices fill uniquely. For each $k \geq 2$, there is an inner horn on a tree with $k + 1$ vertices whose restriction has cokernel free of rank $k!$. For $k = 1$, every inner horn fills uniquely.*

Proof. For an ambient index tuple, join two included faces that contain it when a common subface contains it. Compatible data assign one value to each connected component of this graph, whereas a filler assigns one value to all copies of the tuple. Hence a horn fills for all compatible data if every nonempty graph is connected.

Let T have $n \geq 3$ vertices and o outer faces. The outer vertices are the endpoints of the tree of vertices, so $o \geq 2$. The included contractions form a clique among the faces containing a tuple, as do the included prunings. A contraction and a pruning also give adjacent copies when the contracted edge is away from the pruned vertex. In each case the two moves commute, and the common face retains their common colours. If a tuple has a disconnected graph, exactly two included faces retain it. They are the pruning of an outer vertex v and the contraction of its incident inner edge e. Indeed, an outer vertex has only one incident inner edge, and for $n \geq 3$ an inner edge has at most one outer endpoint.

The remaining included faces delete pairwise disjoint sets of edges. The tuple must contain at least one edge from each such set, so it contains at least $n + o - 4$ distinct edges. If $o = 2$, the tree of vertices is a path. Pruning v and then its neighbour, or contracting e and then pruning the merged vertex, gives the same common face. It contains the edges deleted by every other included contraction and, if the other pruning is included, all the edges deleted by that pruning. A disconnected tuple must contain an additional edge outside this common face. In either case it contains at least $n - 1$ distinct edges. Thus $n \geq k + 2$ gives existence. For uniqueness, a coordinate of $K_F(T)$ must hit the deletion sets of all $n + o - 2$ included faces, so $K_F(T) = 0$ if $n > k$.

For sharpness, take $k + 1$ vertices on a path, all unary except the second vertex from the top, which has an extra leaf b. Let e be the inner edge incident to the top vertex, and omit contraction of another inner edge h. Use a tuple consisting of the root edge, b, and every inner edge except e and h. Only contraction of e and pruning of the top vertex retain this tuple. On contracting e an operation using the merged vertex has the top leaf as an input, and that leaf is absent from the pruned tree. On pruning the top vertex an operation using the second vertex has e as an input, and e is absent from the contracted tree. The colour b is therefore isolated from the root in their common subfaces. The two copies of the tuple may take distinct values, which prevents a filler. For this tree the obstructing tuples are exactly the permutations of the one just described. At the top endpoint the forced inner edges and the root require one further colour, which must be b. At the root endpoint, if its incident contraction is included, the common face contains every shared colour, since the neighbour of the root vertex is unary. The cokernel of the horn restriction is therefore free of rank $k!$.

For one axis the bound covers all trees with at least three vertices. On a two-vertex tree the two faces in the inner horn meet in their connecting edge, so the horn fills by gluing. Their deletion sets give uniqueness. □

A Kernel families over an infinite field

Assume now that K is infinite. Retain the notation P, C, g_T, and $L(T)$ from Section 6. We parameterize the kernels on the locus where their dimensions are minimal.

For a finite-dimensional K-vector space V and $0 \leq r \leq \dim V$, write $\mathrm{Gr}(r, V)$ for the Grassmannian of r-dimensional linear subspaces of V. This is a projective K-variety whose points over a field

extension F/K are the r-dimensional F-linear subspaces of $V \otimes_K F$. Its tautological subbundle of the trivial bundle with fiber V has fiber L at the point representing $L \subseteq V$.

View $W = X_n$ as an affine space. For $0 \le p \le n$, let r_p be the maximal rank of $g_{T,p}$ and put $\ell_p = \dim C_p - r_p$. The locus $U \subset W$ on which all these ranks are attained is a nonempty open subset. Constant-rank kernels define a morphism

$$\Psi : U \longrightarrow G = \prod_{p=0}^{n} \mathrm{Gr}(\ell_p, C_p), \qquad T \longmapsto (L_p(T))_p. \tag{10}$$

The tuples of kernels are chain subcomplexes of C, since g_T is a chain morphism. Chevalley's theorem [12, Tag 054K] makes the image $\mathcal{M} = \Psi(U)$ constructible. Let Y be its reduced Zariski closure. We prove that Y is rational over K, that is, birational over K to an affine space.

Proposition A.1. *The reduced closure Y of the kernel image is rational over K.*

Proof. The closure is integral because U is irreducible. Each $g_{T,p}$ depends linearly on T. For a tuple $L = (L_p)$, the conditions

$$g_{T,p}(L_p) = 0 \quad (0 \le p \le n)$$

therefore define a linear subspace $V_L \subset W$. Let $\mathcal{E} \subset W \times Y$ be the incidence scheme cut out by these conditions on the tautological subbundles L_p pulled back from the factors of G. Over $U \times Y$, L_p and $\ker g_{T,p}$ are subbundles of equal rank, so inclusion forces equality. Hence $\mathcal{E} \cap (U \times Y)$ is the graph of Ψ as a scheme, and its fiber over L is $U \cap V_L$.

Let $F = K(Y)$ be the function field of Y, let η be its generic point, and write W_F, U_F for base change from K to F. Let $V_\eta \subset W_F$ be the linear subspace defined by the incidence conditions at η. Its intersection with U_F is nonempty, since Ψ is dominant onto its image closure. Put $d = \dim_F V_\eta$. Choose d coordinate functions on W whose restriction gives an isomorphism $V_\eta \to F^d$, and denote their joint map by $\pi : W \to K^d$. Solving the defining linear equations expresses the unique element of V_η with $\pi(T) = c$ rationally in the kernel coordinates and linearly in c. The condition that this element belongs to U_F defines a nonempty open subset of F^d. Since K is infinite, K^d is Zariski dense in F^d; choose $c \in K^d$ in that open subset.

After restricting to a nonempty open subset of $\overline{\mathcal{M}}$, the subspaces V_L have constant dimension, $\pi|_{V_L}$ is invertible, and the solutions with $\pi(T) = c$ belong to U. Every fiber there consequently meets the affine slice $S = \pi^{-1}(c)$ in exactly one point, varying rationally with L. These solutions and $\Psi|_S$ are inverse on nonempty open subsets. As $S \cong \mathbb{A}_K^{\dim W - d}$, the closure is rational. □

For $V = K^3$, a plane is the kernel of a nonzero linear form (a, b, c), unique up to a nonzero scalar. Thus

$$\mathrm{Gr}(2, K^3) \xrightarrow{\sim} \mathbb{P}^2_K, \qquad \ker(a, b, c) \longmapsto [a : b : c],$$

where $[a : b : c]$ denotes the corresponding homogeneous coordinates.

Proposition A.2. *For shape $(3, 3)$, the generic normalized kernel has dimensions $2, 0, 0$ in degrees $0, 1, 2$. Under the displayed Grassmannian identification, the kernel image is*

$$\mathcal{M} = \mathbb{P}^2_K \setminus \{[1 : 1 : 1]\}.$$

For every $T \in U$,

$$H_1(\langle T \rangle) \cong K, \qquad H_j(\langle T \rangle) = 0 \quad (j \ne 1).$$

Proof. Write $T = (t_{ij})$ and $(a, b, c) = (t_{00}, t_{11}, t_{22})$. In degree zero, f_T is the row $(a, b, c) : K^3 \to K$. In degree one, the columns corresponding to the three constant maps $[1] \to [2]$ are aJ, bJ, cJ, where J is the all-ones 2×2 matrix. When $(a, b, c) \neq 0$, they span KJ. Modulo this line, the other three columns are represented by

$$B_T = \begin{pmatrix} t_{01} - a & t_{02} - a & t_{12} - b \\ t_{10} - a & t_{20} - a & t_{21} - b \\ b - a & c - a & c - b \end{pmatrix}.$$

Its first two rows are arbitrary, while its last row is nonzero exactly when a, b, c are not all equal. Consequently $f_{T,1}$ can have rank 4 for a specified diagonal precisely when that diagonal is nonconstant. Such a choice gives $\operatorname{rank} g_{T,0} = 1$ and $\operatorname{rank} g_{T,1} = 3$, using the natural decomposition $P_1 = C_1 \oplus s_0 C_0$ and its counterpart for X.

The normalized simplex complex $C = N(K[\Delta^2])$ has dimensions $(3, 3, 1)$ and injective differential $C_2 \to C_1$, since its two-dimensional simplex has nonzero boundary. Since $g_{T,1}$ is injective, the chain-map identity makes $g_{T,2}$ injective as well. These are maximal ranks in all degrees, and $U = \{T : \det B_T \neq 0\}$. Thus on U the kernel complex is $\ker(a, b, c)$ in degree zero and zero elsewhere. Conversely, every nonconstant projective diagonal is realized by choosing the first two rows of the displayed matrix to complete its last row to a basis. This construction works over every residue field. Under the displayed identification of $\mathrm{Gr}(2, K^3)$ with $\mathbb{P}^2_K$, the kernel map therefore has exactly the stated image.

The restriction of the augmentation $(1, 1, 1)$ to $\ker(a, b, c)$ is surjective with one-dimensional kernel, since (a, b, c) is nonconstant. Formula (6) proves the homology assertions. These quotients have the same chain dimensions and differential ranks. By (7) they are isomorphic, and Proposition 6.1 places their kernels in one automorphism orbit. □

For comparison, the all-ones 3×3 tensor generates the constant simplicial module K: every simplicial operator sends it to an all-ones tensor. Its homology is K in degree zero and zero in positive degrees. Thus generated submodules distinguish tensors by homology even when their common ambient module is contractible.

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